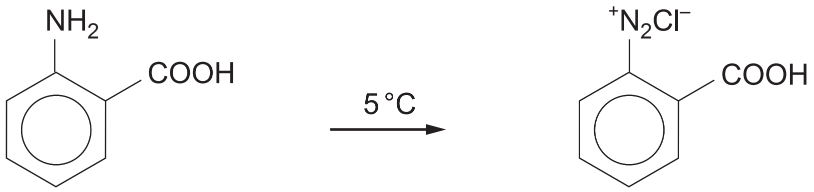
Examiner only

10. *(a)* The acid-base indicator methyl red can be made from 2-aminobenzenecarboxylic acid.

(i) In the first stage of the reaction, a benzenediazonium compound is made from this acid.

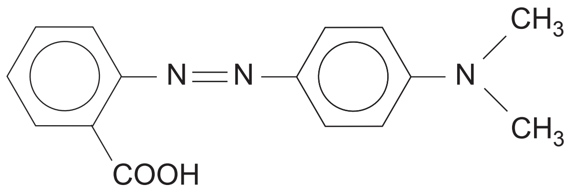


2-aminobenzenecarboxylic acid

State the reagent(s) necessary to produce this diazonium compound from the acid. [1]

....

(ii) The benzenediazonium compound is then coupled with an amine to make methyl red.



methyl red

Draw the structure of the amine used. [1]

(iii) The visible spectrum of this azo dye at a certain pH shows an absorption maximum at 410 nm.

Calculate the frequency that corresponds to this wavelength. [2]

Frequency = ... Hz

Examiner only

(b) Some azo dyes are permitted for use as food colours but their concentration is strictly controlled. The concentration present can be found by colorimetry. The absorption of a series of standard solutions is measured and a straight line calibration graph is drawn.

(i) The molar mass of an azo dye is 272 g mol^{-1}.

Calculate the mass of the azo dye needed to prepare 250 cm^3 of a standard solution of concentration 0.0075 mol dm^{-3}. [2]

Mass required = ... g

(ii) Absorption (A) and concentration (c) are related by the equation

$A = k\,c$ where k is a constant

The calibration graph shows that an absorption of 1.44 is measured for a solution of concentration 0.0096 mol dm^{-3}.

Calculate the concentration of a solution that gave an absorption of 1.03. [2]

Concentration = ... mol dm^{-3}

(c) (i) When ethanoic acid is heated with urea, NH_2CONH_2, the products are ethanamide, carbon dioxide and ammonia.

Give the equation for this reaction. [1]

....

(ii) Suggest why an amide can act as a base. [1]

....

....

Examiner only

(iii) When ethanamide is heated with a dehydrating agent the organic product is ethanenitrile.

Use the **Data Booklet** to state and explain how characteristic infrared absorption values change during the reaction. In your answer you should relate the relevant bonds to their corresponding absorption frequencies. [2]

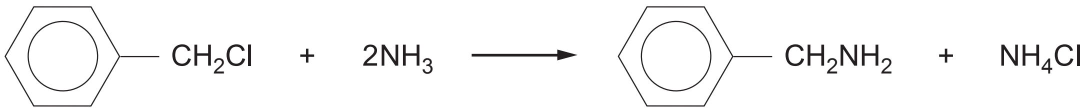
....

....

....

....

(iv) Some amines can be made by the reaction of a chlorine-containing compound and ammonia. For example (chloromethyl)benzene reacts with ammonia to give phenylmethylamine.



Suggest why phenylamine cannot easily be made by this method from chlorobenzene and ammonia. [1]

....

....

....

13